

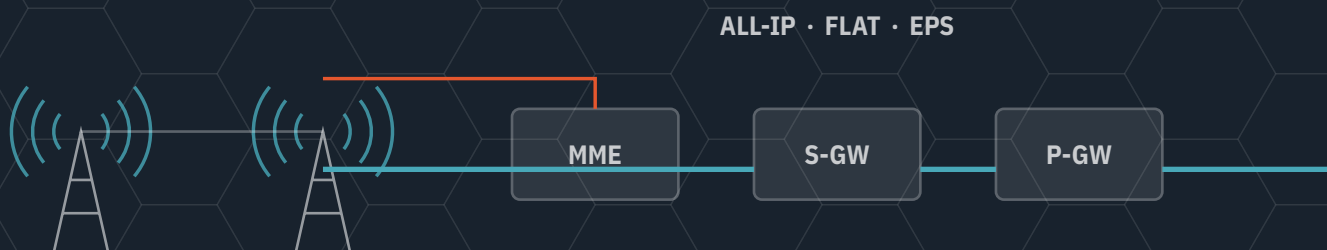
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 5

4G LTE Architecture

معماری 4G LTE



Lecture Notes

Instructor: Amin Azari

Compiled by Aidin Shekari

استاد: امین آذری github.com/AminAzari

In This Module

5.1 Why LTE Was Developed	2
5.2 LTE Design Principles	3
5.3 LTE Architecture Overview	4
5.4 Network Elements — Detailed	6
5.5 LTE Interfaces — Reference Points	12
5.6 IMS Overview (IP Multimedia Subsystem)	18
5.7 Control Plane vs User Plane Separation (CUPS)	20
5.8 Key Differences from 3G	21
5.9 Control Plane vs User Plane Paths	23
5.10 Interface Summary Table	26
Key Takeaways	27
References	27

5.1 Why LTE Was Developed

■ Limitations of 3G (UMTS/HSPA)

The 3rd Generation Partnership Project (3GPP) initiated the Long Term Evolution (LTE) project in 2004 to address fundamental limitations in 3G networks:

Architectural Complexity:

- 3G UTRAN required a hierarchical structure: UE → NodeB → RNC → CN
- The Radio Network Controller (RNC) was a single point of failure and bottleneck
- Complex protocol stacks between NodeB and RNC (Iub interface) added processing overhead
- Soft handover required coordination between multiple NodeBs via the RNC

Latency Issues:

- 3G round-trip times (RTT): 50–100 ms typical, 150+ ms under load
- The RNC added processing hops in both control and user planes
- Circuit-switched voice added further complexity and delay
- Real-time applications (gaming, video calls) suffered noticeably

Spectrum Efficiency:

- WCDMA/HSPA approaching theoretical limits
- Demand for mobile data growing exponentially (smartphones era beginning)
- Need for flexible bandwidth: 1.4 MHz to 20 MHz channel widths

All-IP Target:

- 3G still maintained circuit-switched domain for voice (MSC, MGW)
- Dual-stack (CS + PS) increased OPEX and complexity
- Industry consensus: everything must converge to IP

Performance Targets for LTE (3GPP Release 8):

| Parameter | Target |

|-----|-----|

| Peak DL rate | 100 Mbps (20 MHz) |

| Peak UL rate | 50 Mbps (20 MHz) |

| User plane latency | < 5 ms |

| Control plane latency | < 100 ms (idle→active) |

| Spectrum efficiency | 3–4x improvement over HSPA |

| Mobility | Up to 350 km/h |

5.2 LTE Design Principles

LTE was designed from scratch with these guiding principles:

■ Flat Architecture

- **Eliminate the RNC** — distribute its functions to the eNodeB and core network
- Fewer network nodes = fewer hops = lower latency
- Each eNodeB connects directly to the core network (EPC)

■ All-Packet (No Circuit Switching)

- **Pure packet-switched network** — no circuit-switched domain
- Voice handled via VoLTE (Voice over LTE) using IMS
- Single protocol stack simplifies operations

■ No Soft Handover

- 3G soft handover required simultaneous connections to multiple NodeBs
- LTE uses **hard handover** only — simpler, more spectrum-efficient
- Handover is fast enough (~20–50 ms interruption) to be imperceptible

■ Simplified Protocol Stack

- Fewer layers, cleaner separation of concerns
- User plane: PDCP → RLC → MAC → PHY
- Control plane: NAS (between UE and MME) + RRC (between UE and eNB)

■ Shared Channel Access

- No dedicated channels — all resources shared dynamically
- Scheduler in eNodeB makes 1 ms TTI decisions
- Enables statistical multiplexing gains

■ Self-Organizing Network (SON) Features

- Automatic neighbor relations (ANR)
- Self-configuration of new eNodeBs
- Self-optimization and self-healing capabilities

5.3 LTE Architecture Overview

LTE architecture consists of two main domains:

1. **E-UTRAN** (Evolved UTRAN) — the radio access network
2. **EPC** (Evolved Packet Core) — the core network

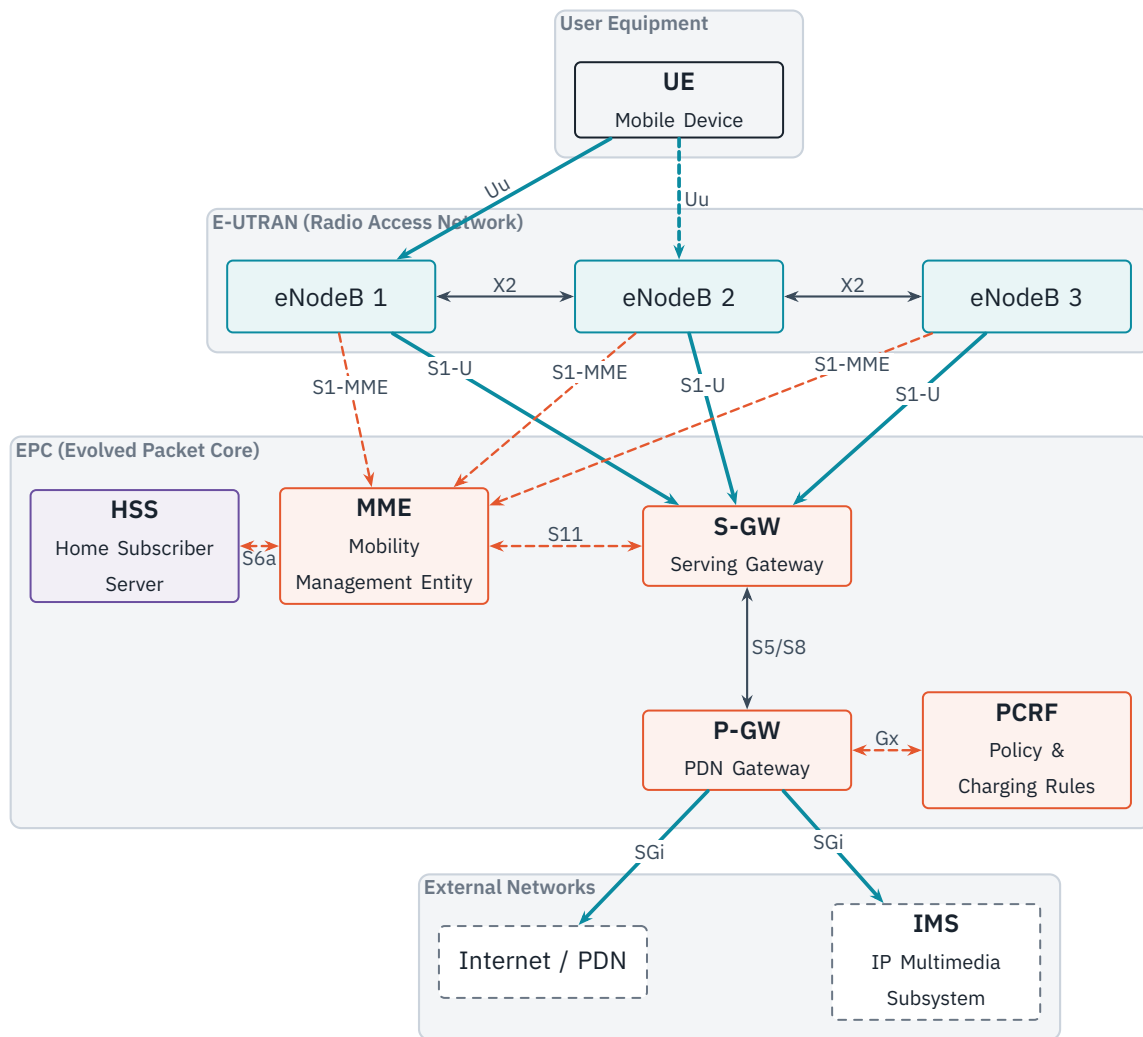


Figure 5.1 LTE architecture: UE, E-UTRAN, EPC and external networks

■ E-UTRAN: The Radio Access Network

Key insight: E-UTRAN consists of eNodeBs only — there is NO RNC!

- The eNodeB handles ALL radio-related functions previously split between NodeB and RNC
- eNodeBs connect directly to the EPC via the S1 interface
- eNodeBs connect to each other via the X2 interface for handover and load balancing
- This is the “flat architecture” — one node type in the RAN

■ EPC: The Evolved Packet Core

The EPC is a pure packet core with separated control and user planes:

Element	Primary Role
MME	Control plane — signaling, mobility, security
S-GW	User plane anchor within E-UTRAN
P-GW	User plane anchor to external networks
HSS	Subscriber database and authentication
PCRF	Policy and charging rules

5.4 Network Elements — Detailed

5.4.1 UE (User Equipment)

Purpose: The mobile device that communicates over the LTE air interface.

Components:

- **ME (Mobile Equipment):** The physical device (phone, tablet, modem)
- **USIM (Universal Subscriber Identity Module):** Stores subscriber identity (IMSI), security keys (K), and authentication algorithms

UE Categories (3GPP Release 8–12):

Category	Max DL (Mbps)	Max UL (Mbps)	DL MIMO	Notes
Cat 1	10	5	None	IoT, basic
Cat 2	51	25	2×2	Entry smartphones
Cat 3	100	50	2×2	Mainstream
Cat 4	150	50	2×2	High-end
Cat 5	300	75	4×4	Flagship
Cat 6	300	50	2×2	Carrier Aggregation
Cat 9	450	50	2×2	3CC CA
Cat 12	600	100	4×4	Advanced
Cat 16	980	150	4×4	LTE-Advanced Pro

Key UE Capabilities:

- Carrier Aggregation (CA): combine multiple component carriers
- MIMO support (2×2 or 4×4 antenna configurations)
- VoLTE support (IMS registration, SIP signaling)
- Dual connectivity (connect to two eNodeBs simultaneously — Release 12+)

5.4.2 eNodeB (Evolved NodeB)

Purpose: The single RAN node that handles ALL radio access functions — replaces both NodeB AND RNC from 3G.

Problem it solves: Eliminates the RNC bottleneck, reduces latency by one hop, enables faster scheduling decisions.

Functions absorbed from 3G RNC:

- Radio Resource Management (RRM) — scheduling, admission control
- Radio Bearer Control — setup, maintenance, release of bearers
- Connection Mobility Control — handover decisions
- Dynamic Resource Allocation (Scheduler) — 1 ms TTI granularity
- Inter-cell Interference Coordination (ICIC)
- Header compression (ROHC via PDCP)
- Encryption and integrity protection (PDCP layer)
- Segmentation and reassembly (RLC layer)

Functions unique to eNodeB:

- S1 interface termination (to MME and S-GW)
- X2 interface termination (to neighboring eNodeBs)
- Paging message distribution
- Broadcast information (SIBs)
- Measurement configuration and reporting

Scheduler — The Brain of eNodeB:

- Makes resource allocation decisions every 1 ms (TTI)
- Assigns Resource Blocks (RBs) to UEs based on:
 - Channel quality (CQI reports from UEs)
 - QoS requirements (GBR vs non-GBR bearers)
 - Fairness algorithms (Proportional Fair, Round Robin, Max Throughput)
 - Buffer status and priority

5.4.3 MME (Mobility Management Entity)

Purpose: The control plane anchor of the EPC — handles all signaling and control functions, touches NO user data.

Problem it solves: Separates control from user plane, enabling independent scaling. Centralizes mobility and security management.

Key Functions:

Function	Description
NAS Signaling	Manages Non-Access Stratum protocols (EMM, ESM) between UE and MME
Authentication	Generates authentication vectors with HSS (EPS-AKA)
Security	NAS encryption/integrity; derives keys for AS security
Mobility Management	Tracks UE location (Tracking Areas), handles TAU
Bearer Management	Establishes, modifies, releases EPS bearers
Paging	Initiates paging when downlink data arrives for idle UE
S-GW Selection	Selects appropriate Serving Gateway for UE
P-GW Selection	Selects PDN Gateway (during initial attach or new PDN connection)
Handover Control	Inter-eNB and inter-MME handover signaling
Idle Mode	Manages UE state transitions (connected ↔ idle)

NAS Procedures:

- **Attach:** UE registers with the network (identity, authentication, bearer setup)
- **Detach:** UE deregisters (voluntary or network-initiated)
- **TAU (Tracking Area Update):** UE reports new location when crossing TA boundary
- **Service Request:** Idle UE requests to become active (for data transfer)

5.4.4 S-GW (Serving Gateway)

Purpose: The user plane anchor within the E-UTRAN — all user data passes through S-GW.

Problem it solves: Provides a stable user plane anchor point during intra-LTE mobility (handover between eNodeBs). Only one tunnel switch needed at S-GW during handover instead of rerouting through external networks.

Key Functions:

- **Local mobility anchor:** During inter-eNB handover, only the S1-U tunnel is switched at S-GW
- **Packet routing and forwarding:** Routes packets between eNB and P-GW
- **Data buffering during handover:** Buffers DL packets while handover completes
- **Downlink data notification:** Triggers paging via MME when DL data arrives for idle UE
- **Lawful intercept:** Interface for legal interception of user traffic
- **Charging data collection:** Usage data for offline charging (CDRs)
- **Transport level marking:** DSCP marking for QoS in transport network
- **Idle mode buffering:** Buffers first DL packet and notifies MME for paging

Deployment Note: In many networks, S-GW and P-GW are combined into a single node (SAE-GW) for simplicity.

5.4.5 P-GW (PDN Gateway)

Purpose: The point of connection to external packet data networks (Internet, IMS, enterprise VPNs). Allocates IP addresses and enforces policy.

Problem it solves: Provides a single, stable IP anchor point for the UE regardless of mobility. The UE keeps the same IP address even when moving between S-GWs.

Key Functions:

- **IP address allocation:** Assigns IPv4 and/or IPv6 addresses to UEs (via DHCPv4, SLAAC, or static)
- **Policy enforcement:** Applies QoS rules received from PCRF (via Gx interface)
- **Charging:** Online charging (Gy to OCS) and offline charging (Gz to OFCS)
- **Packet filtering:** Deep packet inspection, TFT (Traffic Flow Template) enforcement
- **Per-SDF (Service Data Flow) treatment:** Apply different QoS/charging per traffic flow
- **Inter-operator mobility anchor:** When roaming, P-GW remains in home network (home-routed) or visited network (local breakout)
- **NAT (if needed):** Network Address Translation for IPv4
- **GTP/PMIP anchor:** Terminates S5/S8 tunnel from S-GW

APN (Access Point Name):

- Each PDN connection is identified by an APN
- P-GW selection based on APN (e.g., “internet”, “ims”, “enterprise.vpn”)
- Multiple simultaneous PDN connections possible (e.g., Internet + IMS for VoLTE)

5.4.6 HSS (Home Subscriber Server)

Purpose: The central database containing all subscriber information and security credentials. Evolution of 3G HLR/AuC.

Problem it solves: Centralizes subscriber management, provides authentication vectors, and stores subscription data (allowed APNs, QoS profiles, roaming permissions).

Key Functions:

- **Subscriber data storage:**
 - IMSI (permanent identity)
 - MSISDN (phone number)
 - Subscribed APNs and associated QoS profiles
 - Roaming restrictions
 - Subscribed Tracking Areas
- **Authentication vector generation:**
 - Stores permanent key K (shared with USIM)
 - Generates authentication vectors (RAND, AUTN, XRES, K_{ASME}) using EPS-AKA
 - Provides vectors to MME for UE authentication
- **Location management:**
 - Stores current serving MME for each subscriber
 - Used for terminating calls/sessions (where to page the UE)
- **Authorization:**
 - Determines which services a subscriber can access
 - Enforces subscription limits (max bitrate, allowed APNs)

Interfaces:

- S6a to MME (Diameter protocol)
- Cx/Dx to IMS (for VoLTE subscriber data)
- Sh to Application Servers

5.4.7 PCRF (Policy and Charging Rules Function)

Purpose: The policy brain of the network — makes real-time decisions about QoS and charging rules applied to each data flow.

Problem it solves: Enables dynamic, per-flow QoS (e.g., guarantee bandwidth for a VoLTE call while best-effort for web browsing). Enables flexible charging models (zero-rating, tiered plans, sponsored data).

Key Functions:

- **Policy decision:** Determines QoS parameters (QCI, ARP, MBR, GBR) for each service data flow
- **Charging rules:** Specifies how each flow is charged (rating group, metering method)

- **PCC (Policy and Charging Control) rules:** Combined policy + charging instructions sent to P-GW
- **Dynamic policy updates:** Can modify rules in real-time (e.g., when user starts a video call)
- **Subscription-based policy:** Considers user's subscription tier
- **Network-condition-based policy:** Can throttle based on congestion

Example PCC Rule:

```
Rule: "VoLTE_Voice"
- Service Data Flow: SIP/RTP traffic to IMS
- QCI: 1 (Conversational Voice)
- GBR: 40 kbps UL / 40 kbps DL
- ARP: Priority 1, pre-emption capable
- Charging: Included in voice plan (no data deduction)
```

Interfaces:

- Gx to P-GW (installs/modifies PCC rules)
- Rx from IMS/AF (receives session info, triggers policy)
- Sp to SPR (Subscription Profile Repository)

5.5 LTE Interfaces – Reference Points

Interface Map

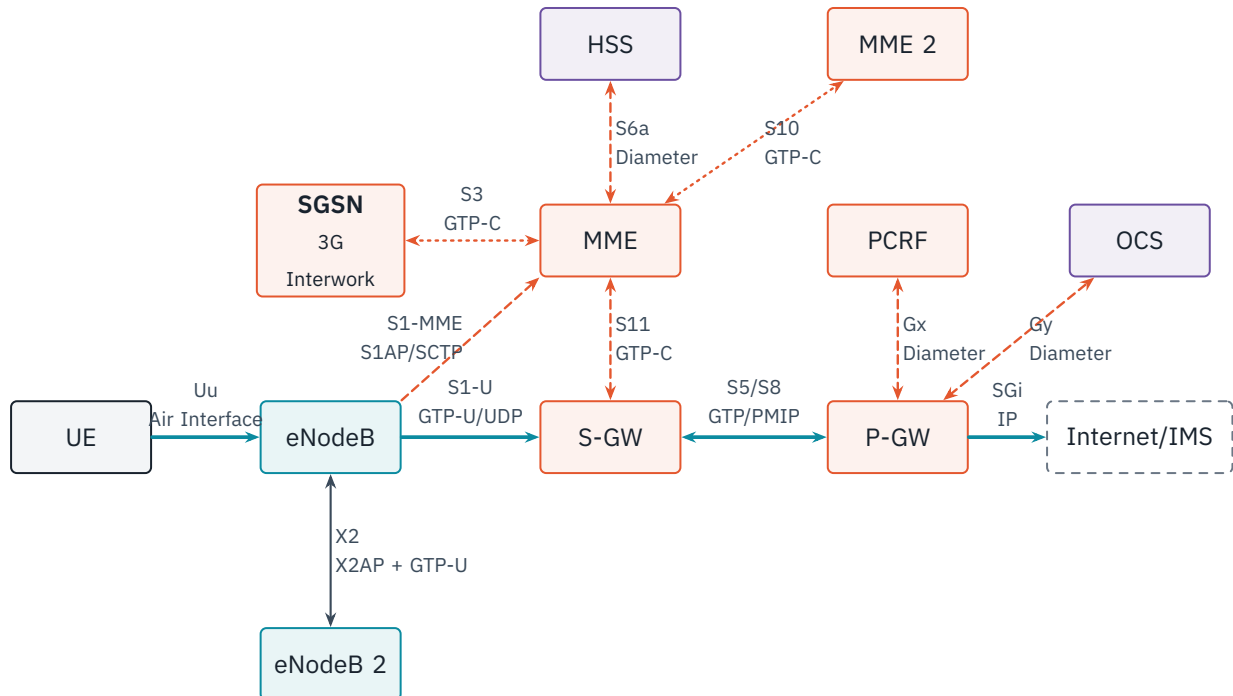


Figure 5.2 LTE interface map with the protocol of each reference point

5.5.1 S1-MME Interface

Property	Value
Endpoints	eNodeB ↔ MME
Plane	Control Plane
Protocol Stack	S1AP / SCTP / IP
Purpose	All control signaling between RAN and core

What flows over S1-MME:

- Initial UE messages (Attach, TAU, Service Request)
- UE context setup/release
- Handover preparation and execution (inter-eNB via MME)
- Paging requests from MME to eNB
- Bearer setup/modification/release commands
- NAS message transport (MME↔UE via eNB as relay)
- Reset and error indication procedures

Why SCTP (not TCP):

- Multi-homing: survives link failures without session drop
- Multi-streaming: avoids head-of-line blocking
- Message-oriented (not byte-stream): natural fit for signaling
- Built-in heartbeat for connection monitoring

5.5.2 S1-U Interface

Property	Value
Endpoints	eNodeB ↔ S-GW
Plane	User Plane
Protocol Stack	GTP-U / UDP / IP
Purpose	Carries user IP packets in GTP tunnels

What flows over S1-U:

- All user data (encapsulated in GTP-U tunnels)
- One GTP tunnel per EPS bearer per UE
- End marker packets during handover (signals last packet from old path)

GTP-U Tunnel Identification:

- Each tunnel identified by TEID (Tunnel Endpoint Identifier) + IP address
- TEIDs allocated independently at each end
- Enables multiplexing many UE bearers over single transport connection

Why GTP-U (not plain IP):

- Tunneling allows mobility transparency (IP address doesn't change)
- Per-bearer QoS enforcement
- Simple encapsulation: GTP-U header (8+ bytes) + inner IP packet

5.5.3 X2 Interface

Property	Value
Endpoints	eNodeB ↔ eNodeB
Plane	Control + User Plane
Protocol Stack	X2AP/SCTP (control) + GTP-U/UDP (user)
Purpose	Direct inter-eNB handover and load management

Control Plane (X2AP) Functions:

- Handover preparation (request, acknowledge, cancel)
- SN (Sequence Number) status transfer
- Load indication (exchange load information between eNBs)

- Resource status reporting
- Inter-cell interference coordination (ICIC) information exchange

User Plane (GTP-U) Functions:

- Forward DL data from source eNB to target eNB during handover
- Prevents data loss during handover gap
- Temporary tunnel — torn down after handover completes

X2 Handover Advantage:

- Direct eNB-to-eNB signaling — no need to involve MME for preparation
- Only MME notified after handover completes (path switch)
- Faster than S1-based handover (used when X2 is available)

5.5.4 S5/S8 Interface

Property	Value
Endpoints	S-GW ↔ P-GW
Plane	User + Control Plane
Protocol Stack	GTP-C + GTP-U (or PMIP alternative)
Purpose	Connects serving gateway to PDN gateway

S5 vs S8:

- **S5**: When S-GW and P-GW are in the **same** PLMN (non-roaming or home-routed)
- **S8**: When S-GW and P-GW are in **different** PLMNs (roaming scenario)
- Technically identical protocol, different name indicates inter-PLMN boundary

Functions:

- Tunnel management (create, update, delete sessions)
- Per-bearer GTP tunnels for user plane
- Triggered by MME (via S11) during attach or bearer modification
- GTP-C handles session/bearer management signaling
- GTP-U carries encapsulated user data

Protocol Options:

- **GTP-based S5/S8**: Full GTP-C signaling + GTP-U tunnels (most common)
- **PMIP-based S5/S8**: Proxy Mobile IPv6 signaling (rare in practice)

5.5.5 S6a Interface

Property	Value
Endpoints	MME ↔ HSS
Plane	Control Plane
Protocol Stack	Diameter / SCTP (or TCP) / IP
Purpose	Subscriber authentication and data retrieval

Diameter Commands on S6a:

Command	Direction	Purpose
Authentication-Information-Request (AIR)	MME→HSS	Request auth vectors
Authentication-Information-Answer (AIA)	HSS→MME	Return auth vectors (RAND, AUTN, XRES, K_{ASME})
Update-Location-Request (ULR)	MME→HSS	Register MME as serving node
Update-Location-Answer (ULA)	HSS→MME	Return subscription data
Cancel-Location-Request (CLR)	HSS→MME	Deregister UE from old MME
Insert-Subscriber-Data-Request (IDR)	HSS→MME	Push updated subscription data
Delete-Subscriber-Data-Request (DSR)	HSS→MME	Remove subscription data
Purge-UE-Request (PUR)	MME→HSS	Inform HSS that UE is unreachable

Why Diameter (not SS7/MAP):

- IP-native (aligns with all-IP architecture)
- Extensible via AVPs (Attribute-Value Pairs)
- Better security (TLS/IPsec support)
- Larger message sizes (supports complex subscription data)

5.5.6 S11 Interface

Property	Value
Endpoints	MME ↔ S-GW
Plane	Control Plane
Protocol Stack	GTP-C v2 / UDP / IP
Purpose	EPS bearer/session management between MME and S-GW

Key Messages:

- **Create Session Request/Response:** During attach — establishes default bearer
- **Modify Bearer Request/Response:** During handover — updates S1-U tunnel info
- **Delete Session Request/Response:** During detach — tears down bearers
- **Create Bearer Request/Response:** MME-initiated dedicated bearer setup
- **Release Access Bearers:** When UE goes idle (release S1-U but keep S5/S8)
- **Downlink Data Notification:** S-GW tells MME that data arrived for idle UE → triggers paging

Why GTP-C v2:

- Already proven in GPRS/3G (GTP-C v1)
- Efficient binary encoding
- Built-in TEID-based message routing
- Supports piggybacking (multiple IEs in one message)

5.5.7 SGi Interface

Property	Value
Endpoints	P-GW ↔ External Networks (Internet, IMS, Enterprise)
Plane	User Plane
Protocol Stack	IP (standard Internet protocols)
Purpose	Connects LTE network to the outside world

Characteristics:

- Equivalent to the Gi interface in 2G/3G GPRS
- Standard IP interface — no GTP or mobile-specific protocols
- P-GW performs NAT here (if needed) or routes native IPv6
- This is where the UE's IP address is “visible” to the internet
- Firewall, DPI, and value-added services typically deployed here

Connected Networks:

- Public Internet
- IMS core (for VoLTE, VoWiFi)

- Enterprise VPNs (via APN-based routing)
- Content Delivery Networks (CDNs)
- Walled garden services

5.5.8 Gx Interface

Property	Value
Endpoints	PCRF ↔ P-GW
Plane	Control Plane
Protocol Stack	Diameter / SCTP or TCP / IP
Purpose	Install, modify, remove PCC (Policy and Charging Control) rules

How it works:

1. P-GW contacts PCRF when a new IP-CAN session is established (UE attaches)
2. PCRF evaluates subscription data + network policy + application input
3. PCRF sends PCC rules to P-GW specifying per-flow QoS and charging
4. PCRF can push updated rules at any time (e.g., when VoLTE call starts)

Diameter Commands:

- **CC-Request/Answer (CCR/CCA):** P-GW reports events, PCRF responds with rules
- **Re-Auth-Request/Answer (RAR/RAA):** PCRF pushes new rules to P-GW

PCC Rule Contents:

- Service Data Flow (SDF) filter (5-tuple: src/dst IP, ports, protocol)
- QoS parameters (QCI, ARP, MBR, GBR)
- Charging key (rating group)
- Metering method (volume, time, event)
- Gate status (open/close a flow)

5.5.9 Gy Interface

Property	Value
Endpoints	P-GW ↔ OCS (Online Charging System)
Plane	Control Plane (charging signaling)
Protocol Stack	Diameter / SCTP or TCP / IP
Purpose	Real-time credit control for prepaid/online charging

How it works:

1. When a prepaid user starts a data session, P-GW requests quota from OCS
2. OCS grants units (bytes, seconds, or events)
3. P-GW meters usage against granted quota
4. When quota runs low, P-GW requests more (or session is terminated)
5. Enables real-time balance deduction and spend limits

Diameter Commands:

- **Credit-Control-Request (CCR):** P-GW requests/reports usage
 - INITIAL: Session start, request first quota
 - UPDATE: Quota running low, request more
 - TERMINATION: Session end, report final usage
- **Credit-Control-Answer (CCA):** OCS grants quota or denies

Use Cases:

- Prepaid data plans with real-time balance
- Fair usage policies (throttle after threshold)
- Sponsored data (third party pays)
- Zero-rating specific applications

5.6 IMS Overview (IP Multimedia Subsystem)

Why VoLTE Needs IMS

LTE is a **pure packet network** — there is no circuit-switched domain for voice. Without IMS:

- Voice calls would fall back to 2G/3G (CSFB — Circuit-Switched Fallback)
- CSFB adds 2–4 seconds call setup delay and drops LTE data temporarily
- No way to provide rich communication services (video calls, messaging)

IMS provides the session control layer that makes VoLTE possible.

IMS Basic Architecture

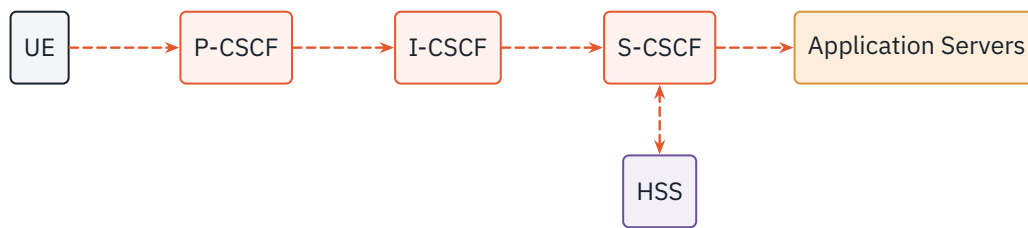


Figure 5.3 IMS signalling path from the UE to the application servers

Key IMS Elements:

Element	Role
P-CSCF	First contact point; SIP proxy in visited network; provides security (IPsec)
I-CSCF	Entry point in home network; queries HSS for S-CSCF assignment
S-CSCF	Core SIP registrar and session controller; applies service logic
Application Servers	VoLTE TAS (Telephony AS), conference, messaging services
MGCF/MGW	Media Gateway — interworks with PSTN for legacy calls

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VoLTE Call Flow (Simplified)

1. UE registers with IMS via SIP REGISTER (through P-CSCF → S-CSCF)
2. UE initiates call via SIP INVITE
3. IMS signals PCRF (via Rx interface) to request dedicated bearer
4. PCRF instructs P-GW (via Gx) to create GBR bearer (QCI=1 for voice)
5. MME/eNB establish dedicated bearer with guaranteed bitrate
6. RTP media flows over the dedicated bearer
7. Result: HD voice with guaranteed QoS, ~100 ms end-to-end

IMS Integration with LTE

- P-CSCF discovered by UE during PDN connection to IMS APN
- UE maintains two PDN connections: Internet APN + IMS APN
- Rx interface: IMS → PCRF (triggers dedicated bearer for voice/video)
- VoLTE requires: UE support + IMS core + PCRF + dedicated bearers

5.7 Control Plane vs User Plane Separation (CUPS)

The Concept

CUPS (3GPP Release 14, TS 23.214) separates the control and user plane functions of S-GW and P-GW into independent entities that can be scaled and deployed independently.

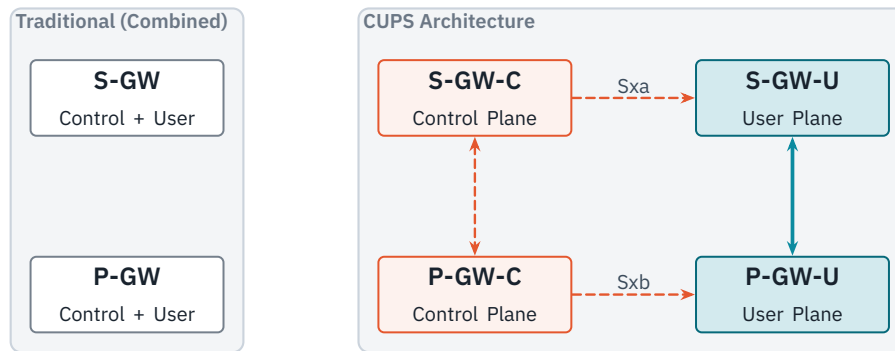


Figure 5.4 Combined gateways versus CUPS (control/user plane separation)

Why CUPS Matters

Aspect	Without CUPS	With CUPS
Scaling	Scale entire gateway (wasteful)	Scale UP and CP independently
Deployment	GW in central DC only	UP at edge, CP centralized
Latency	All traffic through central GW	UP at edge = lower latency
Cost	Expensive combined HW	Cheaper, commodity UP hardware
Evolution	Monolithic	Direct path to 5G UPF architecture

CUPS Interface: Sx (PFCP)

- Protocol: PFCP (Packet Forwarding Control Protocol)
- Sxa: SGW-C ↔ SGW-U
- Sxb: PGW-C ↔ PGW-U
- Sxc: TDF-C ↔ TDF-U
- CP installs forwarding rules (PDRs, FARs, QERs, URRs) into UP
- UP executes packet forwarding based on installed rules

CUPS as Bridge to 5G

CUPS is the architectural precursor to 5G's fully separated architecture:

- SGW-C + PGW-C → 5G SMF (Session Management Function)
- SGW-U + PGW-U → 5G UPF (User Plane Function)

- PFCP protocol reused in 5G (N4 interface = Sx evolution)

5.8 Key Differences from 3G

Aspect	3G UMTS/HSPA	4G LTE
RAN Architecture	Hierarchical: NodeB → RNC	Flat: eNodeB only
RAN Controller	RNC (centralized)	None — functions in eNodeB
Core Network	CS domain + PS domain	EPC (PS only, all-IP)
Voice	Circuit-switched (MSC)	VoLTE via IMS (packet)
Handover	Soft handover (macro diversity)	Hard handover only
Air Interface	WCDMA / HSPA	OFDMA (DL) / SC-FDMA (UL)
Channel Type	Dedicated + shared	All shared (no dedicated channels)
Scheduling	2 ms TTI (HSPA)	1 ms TTI
Bearer Concept	RAB + PDP Context	EPS Bearer (unified)
Max Bandwidth	5 MHz fixed	1.4–20 MHz flexible
Peak Rate	42 Mbps (HSPA+)	300+ Mbps (Cat 5+)
Latency (RTT)	50–100 ms	10–20 ms
Duplexing	FDD only (most)	FDD + TDD
Mobility Anchor	SGSN (control+user)	MME (control) + S-GW (user) — separated
Subscriber DB	HLR (SS7/MAP)	HSS (Diameter)
Policy Control	Limited (PCRF later addition)	Native PCRF (PCC from day one)

Architecture Comparison Diagram

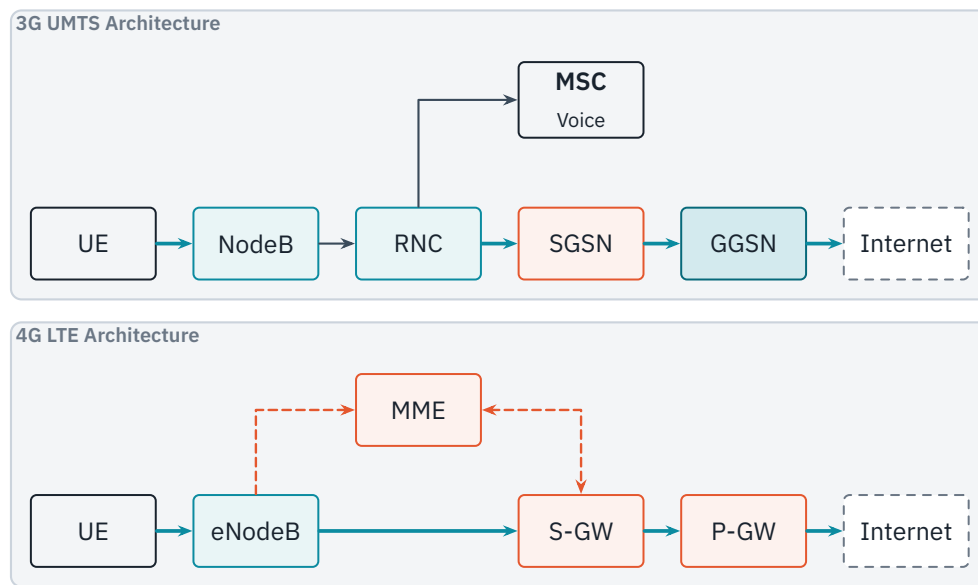


Figure 5.5 3G UMTS versus 4G LTE architecture

Key Architectural Wins

1. Elimination of RNC:

- Removed single point of failure
- Reduced latency by one hop
- Enabled distributed scheduling (faster response)
- Simplified RAN planning and scaling

2. All-IP / No Circuit Switching:

- Single protocol stack to manage
- Reduced OPEX (no MSC, MGW, VLR infrastructure)
- Statistical multiplexing for voice (VoLTE more efficient)
- Unified charging and policy for all services

3. EPS Bearer Model:

- Unified bearer from UE to P-GW (end-to-end QoS)
- Replaces fragmented 3G model (RAB + PDP Context + separate QoS at each interface)
- Default bearer always on (always connected experience)
- Dedicated bearers for specific QoS needs (VoLTE, video)

4. Separation of Control and User Planes:

- MME handles only signaling (scales with number of UEs)
- S-GW handles only data (scales with traffic volume)
- Independent evolution and scaling

5.9 Control Plane vs User Plane Paths

Control Plane Path

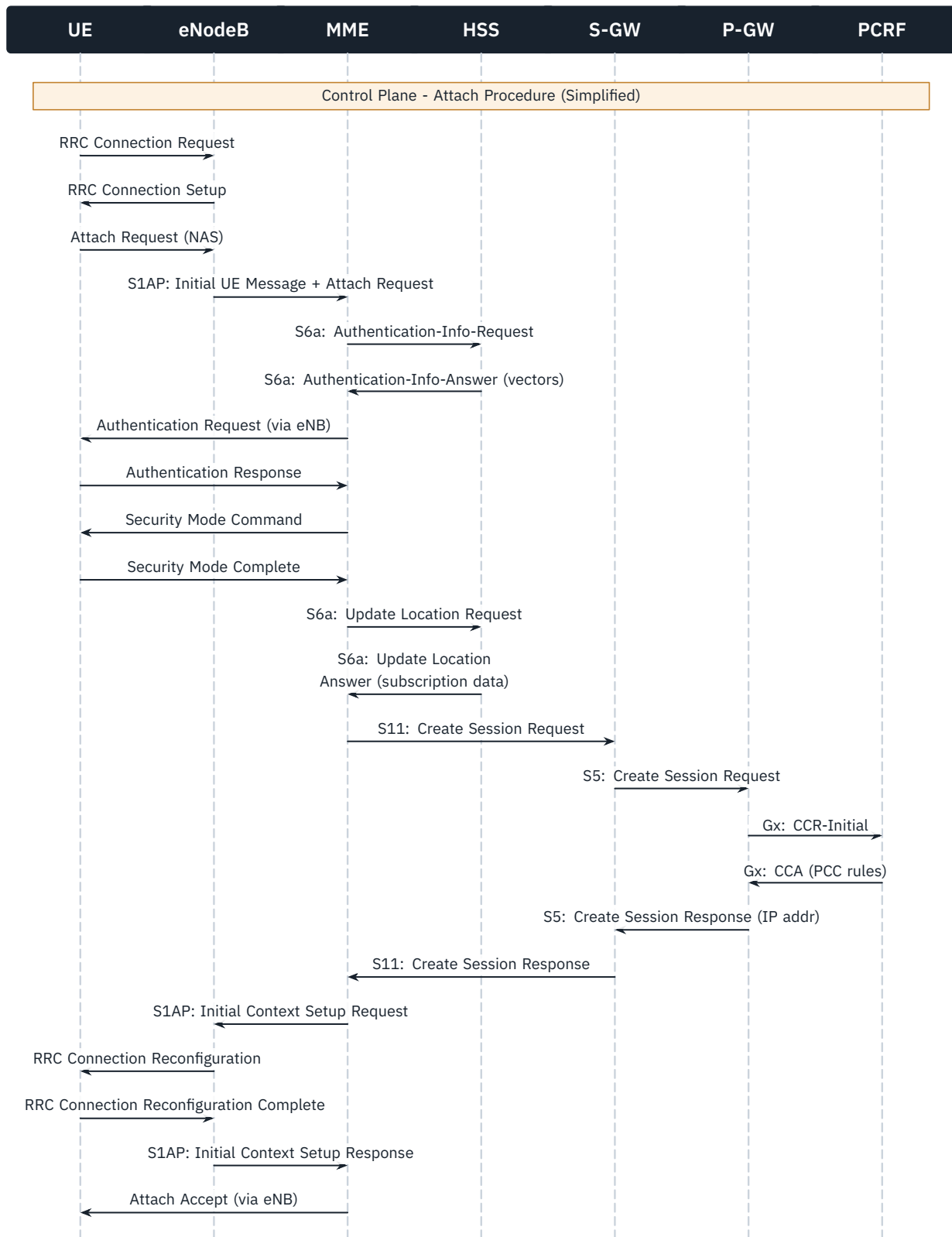


Figure 5.6 Control-plane path: the simplified attach procedure

User Plane Path



Figure 5.7 User-plane path from the UE to the Internet

User Plane Protocol Stack:

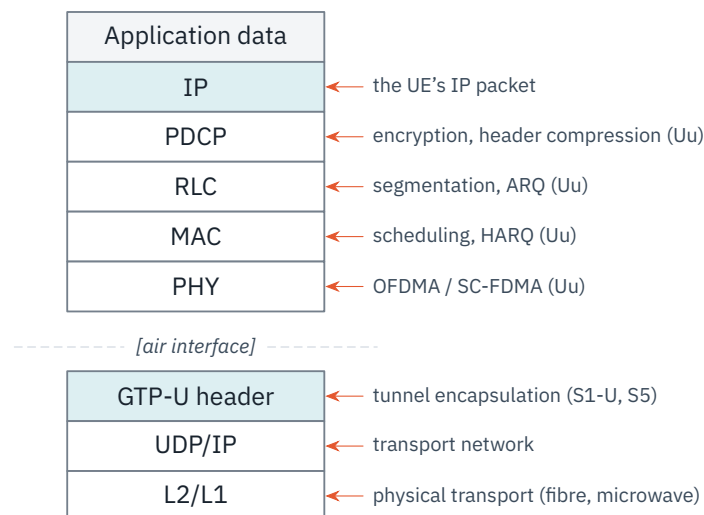


Figure 5.8 User-plane protocol stack of an LTE packet

5.10 Interface Summary Table

Interface	Endpoints	Plane	Protocol	Purpose
Uu	UE ↔ eNodeB	Control + User	LTE-Uu (OFDMA/SC-FDMA)	Air interface — radio communication
S1-MME	eNodeB ↔ MME	Control	S1AP / SCTP	RAN-Core control signaling (NAS relay, handover, bearer mgmt)
S1-U	eNodeB ↔ S-GW	User	GTP-U / UDP / IP	User data tunneling between RAN and core
X2	eNodeB ↔ eNodeB	Control + User	X2AP/SCTP + GTP-U/UDP	Inter-eNB handover signaling and data forwarding
S5/S8	S-GW ↔ P-GW	Control + User	GTP-C + GTP-U (or PMIP)	Session/bearer management and user data between gateways
S6a	MME ↔ HSS	Control	Diameter / SCTP	Authentication vectors and subscriber data retrieval
S11	MME ↔ S-GW	Control	GTP-C v2 / UDP	Bearer/session lifecycle management
SGi	P-GW ↔ Internet/IMS	User	IP (native)	External PDN connectivity — UE traffic exits here
Gx	PCRF ↔ P-GW	Control	Diameter	Policy and charging rule installation/modification
Gy	P-GW ↔ OCS	Control	Diameter	Online/real-time credit control (prepaid charging)
Rx	IMS/AF ↔ PCRF	Control	Diameter	Application-triggered QoS/policy requests
S10	MME ↔ MME	Control	GTP-C v2	Inter-MME handover (UE context transfer)
S3	MME ↔ SGSN	Control	GTP-C v2	3G/4G inter-RAT mobility

Interface	Endpoints	Plane	Protocol	Purpose
SBc	MME ↔ CBC	Control	SBc-AP / SCTP	Cell broadcast (emergency alerts, CMAS/ETWS)

Key Takeaways

1. **LTE = Flat RAN + All-IP Core** — simplicity drives performance
2. **eNodeB is the only RAN node** — absorbed all RNC functions
3. **EPC separates control (MME) from user plane (S-GW/P-GW)** — independent scaling
4. **Every interface has a clear, single purpose** — modular, replaceable
5. **GTP tunnels provide mobility transparency** — IP address never changes regardless of movement
6. **PCRF enables dynamic QoS** — different treatment per application flow
7. **IMS replaces circuit-switched voice** — VoLTE = SIP + dedicated bearer
8. **CUPS is the bridge to 5G** — same separation philosophy, evolved into 5G SBA

References

- 3GPP TS 23.401: “GPRS Enhancements for E-UTRAN Access” (EPC architecture)
- 3GPP TS 36.300: “E-UTRA and E-UTRAN; Overall Description” (RAN architecture)
- 3GPP TS 23.214: “Architecture Enhancements for CUPS”
- 3GPP TS 29.274: “GTPv2-C” (S11, S5/S8 control)
- 3GPP TS 29.281: “GTP-U” (S1-U, S5/S8 user)
- 3GPP TS 36.413: “S1AP” (S1-MME interface)
- 3GPP TS 36.423: “X2AP” (X2 interface)
- 3GPP TS 29.272: “S6a/S6d” (MME-HSS Diameter interface)
- 3GPP TS 29.212: “Gx” (PCRF-PCEF Diameter interface)

Module 5 Complete — Next: Module 6: 5G NR Architecture